

Atena Bahrami

+1 587-566-5728 | bahrami.atena@gmail.com | Toronto, Ontario, Canada | LinkedIn ↗

Summary

PhD candidate in geotechnical engineering (University of Alberta, expected October 2026) with 8+ years of combined industry and research experience spanning experimental rock mechanics, discrete element method (DEM) modeling, and subsurface characterization. Track record of designing large-scale testing programs, building predictive geomechanical frameworks, and applying findings to rock characterization, hydraulic fracturing, and reservoir engineering problems. Previously held industry roles in petrophysics and subsurface modeling at the National Iranian Oil Company (NIOC) and Schlumberger-affiliated operations. Serves as Secretary of the Canadian Geotechnical Society (CGS) Equity, Diversity, and Inclusion (EDI) Committee. Engineer in Training (EIT), APEGA. Fluent in English; developing French proficiency (B1).

Skills

- **Programming & Data Analysis:** Python (Pandas, NumPy, Matplotlib), Gnuplot, Git, Bash
- **Productivity Tools:** Microsoft Office, Google Workspace, LaTeX, Zotero, Mendeley, VS Code
- **Computational & Simulation Tools:** ESyS-Particle, PFC3D, Abaqus, RS2, Techlog, Petrel, Geolog, Blender, SLURM (HPC)
- **Experimental & Image Processing:** ParaView, Digital Image Correlation (DIC: VICSnap, VIC2D, VIC3D), Dragonfly ORS, Metavision SDK
- **Laboratory Expertise:** Uniaxial compressive strength (UCS), triaxial compressive strength (TCS), Brazilian tensile strength (BTS), point load index (PLI), sonic wave velocity (P- and S-wave), permeability, porosity
- **Languages:** English (full professional proficiency), French (intermediate, B1 – currently advancing), Persian (native)

Professional Experience

PhD Candidate, *University of Alberta*

Edmonton, Canada, 2021 – October 2026 (expected)

- Developed a methodology using **irregular rock fragments** to characterize mechanical properties, **removing long-standing constraints on specimen geometry** and enabling testing where core recovery is insufficient.
- Designed and executed **200+** mechanical tests on Sulphur Mountain Formation (**Montney equivalent**) siltstone across multiple sizes, strain rates, and bedding orientations using DIC, establishing the **first comprehensive mechanical database** for this formation to directly inform numerical model calibration.
- Conducted a **Central Composite Design (CCD)** parametric study in ESyS-Particle (DEM) with **14** simulation runs to quantify modulus and strength mismatch effects on fracture paths in composite specimens, enabling **fracture path prediction** in heterogeneous geomaterials relevant to hydraulic fracturing, mining, and underground excavation.
- Performed **3D image segmentation** of rock microstructure and incorporated segmented volumes into PFC3D simulations to assess the effect of structural heterogeneity on measured mechanical properties.
- Collaborated with a team of six to design and validate the **world's smallest triaxial cell** for small-scale specimen testing.
- Presenting part of the dissertation findings at **GeoQuebec 2026** (September, Quebec City).

Geomechanics Researcher, *University of Alberta*

Edmonton, Canada, 2019 – 2021

- Compiled a **comprehensive rock mechanical dataset** for the Duvernay Formation through triaxial, UCS, BTS, and sonic velocity tests to support geomechanical modeling of induced seismicity in the Fox Creek area, Alberta.
- Constructed **1D mechanical earth models (MEM)** for **three** wells in Techlog, integrating laboratory and log data to investigate geomechanical controls on induced seismicity and **identifying key stratigraphic controls on stress distribution**.

Subsurface Modeler, *National Iranian Oil Company (NIOC)*

Ahvaz, Iran, 2017 – 2019

- Performed **well log QC, normalization, and petrophysical interpretation** in Geolog and Techlog, including depth matching, facies classification, and porosity-permeability characterization to inform 3D geological models.
- Developed **static 3D subsurface models** in Petrel through structural interpretation, fault modeling, and geostatistical property population to support field development decisions.
- Generated **volumetric estimates** and reserves calculations to evaluate development scenarios and **optimize packer and liner placement** during drilling.
- Collaborated with multidisciplinary teams to assess **optimal well locations** through integrated subsurface and flow simulation analysis.

Junior Petrophysicist, *Well Services of Iran (Schlumberger Methods)*

Tehran, Iran, 2016 – 2017

- Performed **petrophysical analysis** of well logs using GeoFrame, Interactive Petrophysics, and Techlog to evaluate formation properties.

- Supported **logging tool calibration** to ensure data quality and measurement accuracy across field operations.
- Contributed to the preparation of **logging set proposals** for NIOC client teams.

Seismic Interpreter – Intern, *Research Institute of Petroleum Industry*

Tehran, Iran, 2014

- Performed **seismic interpretation** in Petrel, including horizon picking, fault mapping, and formation top correlation across multiple wells.
- Conducted **time-to-depth conversion** to integrate seismic and well log data for subsurface structural mapping.
- Contributed to **interpretation reviews** and presented results to senior geoscientists.

Leadership & Service

Secretary, Equity, Diversity, and Inclusion (EDI) Committee, *Canadian Geotechnical Society (CGS)*

2026 – Present

- Serve as Secretary of the newly established CGS EDI Committee, managing meeting minutes and supporting committee operations.
- Contributing member of the committee's quarterly webinar series initiative.

Weekly Webinar Organizer, *University of Alberta*

2023 – 2025

- Organized and hosted a **weekly geotechnical seminar series** over two years, sourcing and coordinating speakers from academia and industry, managing scheduling, promotion, platform logistics, and session recording.
- Delivered speaker introductions and maintained consistent **20+ attendee** engagement per session across a graduate student audience.

Education

PhD Candidate, Geotechnical Engineering, *University of Alberta*

Edmonton, Canada, 2021 – 2026

M.Sc., Petroleum Exploration Engineering, *University of Tehran*

Tehran, Iran, 2015 – 2018

B.Sc., Petroleum Engineering (Exploration), *Petroleum University of Technology*

Ahvaz, Iran, 2011 – 2015

Professional Affiliations

- Engineer in Training (EIT), Association of Professional Engineers and Geoscientists of Alberta (APEGA) – 2025 to present
- Member, Canadian Geotechnical Society (CGS) – 2021 to present

Certificates & Awards

- Earle Klohn Graduate Scholarship in Geotechnical Engineering, Klohn Crippen Berger (KCB) – 2025
- RAP Bursary for Graduate Studies, Geological Survey of Canada – 2019
- Gender-Based Analysis Plus (GBA+) Certification, Government of Canada – 2022
- Unconventional Reservoir Geomechanics, Stanford University (Online) – 2019
- Reservoir Geomechanics, Stanford University (Online) – 2015